How-To Library

Informatica

Performance Tuning Guidelines for Databricks for PowerExchange for Cosmos DB SQL API

© Copyright Informatica LLC 2019, 2021. Informatica, the Informatica logo, and PowerExchange are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html

Abstract

When you use PowerExchange for Microsoft Azure Cosmos DB SQL API to read data from or write data to Microsoft Azure Cosmos DB SQL API, multiple factors such as hardware parameters, database parameters, application server parameters, and Informatica mapping parameters impact the adapter performance. You can optimize the performance by tuning these parameters appropriately. This article describes general reference guidelines to help you tune the performance of PowerExchange for Microsoft Azure Cosmos DB SQL API on Databricks.

Supported Versions

- PowerExchange for Microsoft Azure Cosmos DB SQL API 10.2.2

Table of Contents

Overview. 2
Performance Tuning Parameters. 2
Cosmos DB Tuning Tips. 2
Best Practices for Tuning Performance. 3
Tune the Mapping. 4
Ports Precision. 4
Data Type Mapping. 4

Overview

Performance tuning is an iterative process in which you analyze the performance, use guidelines to estimate and define parameters that impact the performance, and monitor and adjust the results as required.

This document describes the key hardware, Cosmos DB, Databricks cluster, and Informatica mapping parameters that you can tune to optimize the performance of PowerExchange for Microsoft Azure Cosmos DB SQL API. It also includes a case study that illustrates the impact that the tuning has on the adapter performance.

Note: The performance testing results listed in this article are based on observations in an internal Informatica environment using data from real-world scenarios. The performance of PowerExchange for Microsoft Azure Cosmos DB SQL API might vary based on individual environments and other parameters even when you use the same data.

Performance Tuning Parameters

You can optimize the performance of Power Exchange for Microsoft Azure Cosmos DB SQL API by tuning the following areas:

- Tune the Azure Cosmos DB
- Best Practices for Tuning Performance
- Tune the Mapping

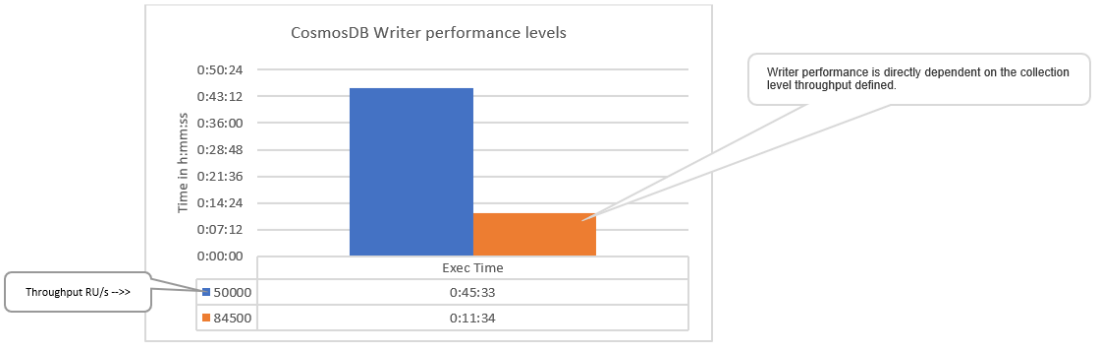
Cosmos DB Tuning Tips

The write or read performance in Azure Cosmos DB is directly proportional to the throughput Resource Units (RUs) set at the collection level

The Azure portal suggests the best throughput based on the volume of data. You must set the throughput accordingly at the data object operation level and set the default throughput RUs at the data object level.

Based on these settings, at the mapping runtime, the throughput at the collection level increases to the value set at the data object operation level and reverts to the default value set at the data object level once the mapping run is complete.

The following image shows the sample performance comparison for the write operation when the throughput is 50000 and 84500:

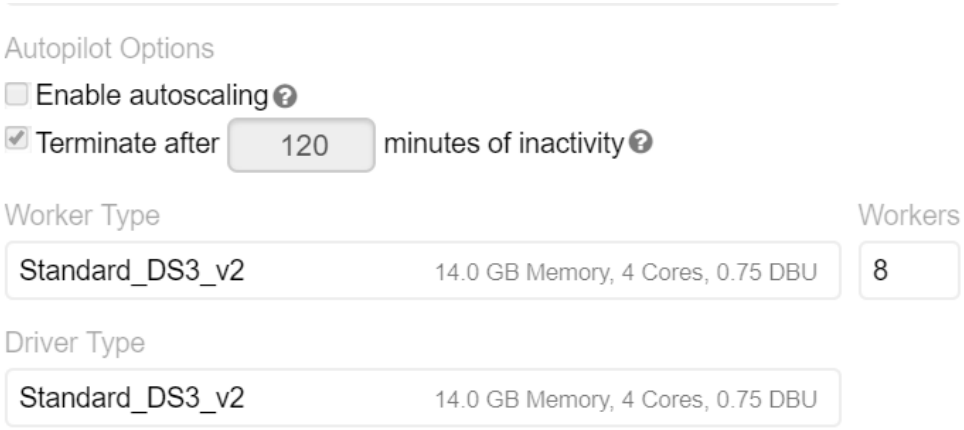


Note: When you increase the throughput, the pricing at the collection level also increases.

Best Practices for Tuning Performance

Consider the following best practices to tune the performance:

- The Domain Host, Source, and Target must be in the same region as that of the Databricks cluster to reduce the network latency and improve the performance.
- Maintain the same datatypes, precision, and scale throughout the mapping to avoid datatype conversions.
- For low Precision data, change the default mode of high precision to low precision to reduce the size of decimal data to be processed by the engine.
- Disable autoscaling and provide maximum number of worker nodes before launching the databricks cluster. The following image describes how to disable autoscaling and provide number of worker nodes before launching the databricks cluster:



- Choose the appropriate instance type for Worker Type and Driver Type, based on the resource requirement. For more information on pricing and resource details, refer to https://azure.microsoft.com/en-in/pricing/details/databricks/

Tune the Mapping

When you run mappings on the Databricks engine, consider the following tuning recommendations and best practices.

Ports Precision

Precision is the maximum number of significant digits for numeric data types, or maximum number of characters for string data types. For numeric data types, precision includes scale.

You can tune the precision in repository mappings.

When mappings contain ports with a larger precision than required, the mapping performance degrades. Informatica recommends that you set the precision judiciously for all source ports, transformation ports, and target ports. For instance, if a string port can handle data of a maximum of 200 characters, set the precision to 200. Do not set the precision to a high value such as 1000.

Data Type Mapping

When the Integration Service reads source data, it converts the native data types to the comparable transformation data types before transforming the data. When the Integration Service writes data to a target, it converts the transformation data types to the comparable native data types.

When you map source ports to transformation ports and then to target ports, avoid unnecessary data type conversions. For instance, do not map a port of the string data type to a port of the date data type. Ensure that you map ports to the same data type in all components of the mapping. Also, remove all unconnected ports from the mapping.

Authors

Krishna Prabhakar Devarakonda

Manav Khurana

Ripa Bhagawati